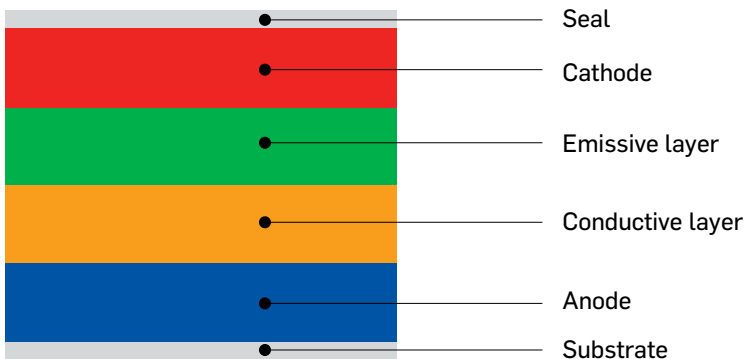


seal and the substrates provide a protective covering and support for the entire assembly. The cathode injects electrons when voltage is applied, whereas the anode removes the electrons. Amongst the organic layers, the conducting layer sends electrons out from the anode whereas the emissive layer transports electrons from the cathode, forming the light and making the OLED glow.

Now, to analyse the working in detail, when a potential difference is created between the anode and the cathode, electricity starts flowing, making the cathode receive electrons and the anode to lose them. The added electrons make the emissive layer negatively charged. Removing electrons from the positively charged conductive layer leave holes that need to be filled in the emissive layer. Positive holes being much more mobile than the negative electrons so the holes jump to the emissive layer and recombine with electrons and thus release the extra energy as light. This process takes place continuously as long as the current flows and thus light is produced in an OLED panel.

The colours that we see i.e red, blue and green are because of a coloured filter added beneath the substrate. A continuous panel filled with thousands of such assembly will give you an OLED display panel that switches these OLEDs on and off independently to make them work as pixels in any conventional LCD screen to produce high definition image.

Currently, Samsung and LG are the two major companies selling OLED TVs with LG having the maximum market share. It is interesting that both the companies have different arrangement methods. On one hand, where Samsung uses the traditional RGB sub-pixel arrangement, LG adds in an



Structure of an OLED panel